

NumPy Problem-Solving Test (Easy → Hard)

- Q1: Create a NumPy array [1,2,3,4,5]. Find its shape and size.
- Q2: Create a 3x3 array of zeros and convert it to float.
- Q3: Generate [0,2,4,6,8] using a NumPy function.
- Q4: Create a 2x3 array filled with 7. Replace element at (1,2) with 99.
- Q5: Given arr=[10,20,30,40,50]: slice index 1 to 3, replace values >30 with 0.
- Q6: Given a=[1,2,3], b=[4,5,6]: multiply element-wise, then take sqrt.
- Q7: Create 4x4 random array. Find mean and column-wise max.
- Q8: Given [[1,2,3],[4,5,6]]: transpose and flatten.
- Q9: Given [2,4,6,8,10]: boolean mask >5, extract and square them.
- Q10: arange(1,13): reshape to 3x4, get 2nd row, last column, subarray (2x2).
- Q11: Concatenate [[1,2],[3,4]] and [[5,6]] row-wise and column-wise.
- Q12: Given [10,25,30,45,50,5]: filter 20–50, replace <20 with -1, find mean & std.